

Markscheme

Mathematics: analysis and approaches
Practice question bank

67. (a)

The sum of the first 12 terms of an arithmetic sequence is 222, and the sum of the first 20 terms is 590.

Find the first term and the common difference.

$$S_{12} = 6(2u_1 + 11d) = 222 \Rightarrow 2u_1 + 11d = 37 \quad (M1) \text{ A1}$$

$$S_{20} = 10(2u_1 + 19d) = 590 \Rightarrow 2u_1 + 19d = 59 \quad (M1) \text{ A1}$$

$$\text{subtract: } 8d = 22 \Rightarrow d = 2.75 \quad (M1)$$

$$u_1 = 3.375 \quad \text{A1}$$

[6 marks]